

Toward Detecting Student-Athletes' Condition Using Passive Mobile and Wearable Sensing

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ABSTRACT

Student-athletes (SAs) face stress from balancing athletic and academic demands; therefore, monitoring their physical and mental stress is crucial to ensure better well-being. Questionnaires and dedicated measurement equipment have been used to assess the mental and physical status of SAs. However, these methods are not scalable and are difficult to use continuously. In this paper, we propose a method for monitoring the physical and mental state of SAs using passive mobile and wearable sensing technology to minimize the burden associated with monitoring. First, we developed a platform for collecting daily, training, and resetting behavior data from smartphones and wearable devices. Second, as a preliminary study, we collected the behavior and Stress and Recovery States Scale (SRSS) data for four weeks from 19 SAs and analyzed the collected data to understand their unique behavior patterns and living environments. The results demonstrate that wearable devices and smartphones can automatically collect data on "exercise intensity during competitions" and "lifestyle patterns," specifically tailored to the mental and physical states of SAs. Furthermore, this preliminary research establishes a foundation for future efforts using machine learning to predict the physical and mental states of SAs.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile computing systems and tools.**

KEYWORDS

Passive Mobile Sensing, Student-Athlete, Sport, Stress, Recovery State

ACM Reference Format:

Zhenbo Wang, Akihito Taya, Takaaki Kato, Kaoru Sezaki, and Yuuki Nishiyama. 2024. Toward Detecting Student-Athletes' Condition Using Passive Mobile and Wearable Sensing. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion)*

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UbiComp Companion '24, October 5–9, 2024, Melbourne, VIC, Australia

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ACM ISBN 979-8-4007-1058-2/24/10

<https://doi.org/10.1145/3675094.3677583>

'24), October 5–9, 2024, Melbourne, VIC, Australia. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3675094.3677583>

1 INTRODUCTION

The mental health of student-athletes (SAs) has been reported to be the same or worse than that of general college students [28]. This paper defines the student-athlete as a student who belongs to an official university athletic team, such as baseball, tennis, and football. Generally, mental health issues among university students are increasingly becoming a significant cause of school dropouts, decreased motivation to learn, self-harm behaviors, and suicide [17]. Specific university life factors, such as poor sleep quality [5], academic stress, and financial burdens [8, 9], can further exacerbate mental health problems. In addition to these stressors, SAs face the unique mental stresses associated with being athletes, making it challenging to maintain good mental health. A survey of 950 collegiate athletes affiliated with the NCAA in the United States revealed that 33.2% exhibited symptoms of depression [6]. The mental health of SA directly impacts their academic performance and increases their risk of physical injury [15, 23], including musculoskeletal injuries [13, 24]. The survey results highlight that balancing academic and athletic activities is a significant source of stress for SAs [7].

Previous research assessing the mental and physical states of athletes has predominantly relied on questionnaires [2, 15, 23, 26] and measurement instruments [13, 24, 25]. These methods are prone to recall bias [3], with participants likely providing socially desirable responses [12]. Additionally, measurement techniques, such as saliva sampling can place a considerable burden on participants [20]. This situation highlights the urgent need for evidence-based and low-burden methods for managing the mental and physical health of SAs.

Smartphones and wearable devices can detect factors related to mental health through real-time changes, thereby overcoming some limitations associated with traditional monitoring methods, and these devices are increasingly being integrated into existing methods to evaluate the mental and physical health of individuals [27]. Furthermore, several studies have demonstrated the effectiveness of mobile applications and wearable devices in reducing stress and improving mental health among university students [16, 22]. However, because current research has mostly focused on university students [29], and the mental health of SAs is closely related to their lifestyle behaviors [4], further investigation is needed to assess the

effectiveness of passive mobile and wearable sensing for SAs, who have different lifestyles.

Long-term observation is crucial for monitoring the diverse lifestyles of SAs. Therefore, this study aims to establish a low-burden monitoring system to promote better long-term observation of SAs' physical and mental conditions. Specifically, passive sensing technology will be employed to gather information related to athletes' actions during training and daily activities, as well as their physical and mental stress and recovery states. The analysis will focus on a foundational investigation of these datasets to understand the performance, daily activities, physiological and psychological stress states, and unique environments of SAs. The contributions of this paper are summarized as follows:

- We designed and developed a system for collecting SAs' behaviors using passive mobile and wearable sensing technologies.
- We created a dataset of SAs' activities, including daily life and training data, based on a data collection study with 19 SA over four weeks.
- Our comprehensive analytics characteristic behavioral tendencies specific to SA and their correlation with the SRSS.

2 RESEARCH QUESTION AND PROCESS DESIGN

Currently, the field of Sports Human-Computer Interaction (SportsHCI) has gained increasing attention, significantly impacting physical and mental health and improving the quality of sports performance [1, 10, 19]. Although passive sensing technology is effective for monitoring the mental health of college students, the unique lifestyle of SAs suggests that these methods might not be directly transferable. This indicates a need for a bespoke data collection systems tailored to their specific environment. Therefore, a comprehensive understanding of the physical and mental conditions of SAs, along with their lifestyle patterns, is necessary.

This study aims to establish an infrastructure to automatically collect sensor data that may affect physical and mental conditions based on passive mobile sensing methods, and detect physical and mental health conditions from the collected data. Fig. 1 shows an overview of our research process. First, a passive data collection system for SAs is designed using consumer smartphones and wearable devices. Through this system, we collected their daily activities and measured their mental and physical conditions using the **Short Recovery and Stress Scale (SRSS)** methods [15]. Then, the features of each sensor dataset were extracted and analyzed in relation to the reference data. Future work will explore the relationship between "behavioral data" and "mental and physical stress, including recovery," aiming to develop a machine learning model to predict psychological and physical states.

3 METHODS

This study collected data over four weeks, from February to March 2023, from 19 male members of a university varsity baseball team ($M = 20.68$, $SD = 1.00$) with a history of winning national championships. The average playing experience was 13.26 years ($SD = 1.94$). The participants' positions included six pitchers, five catchers,

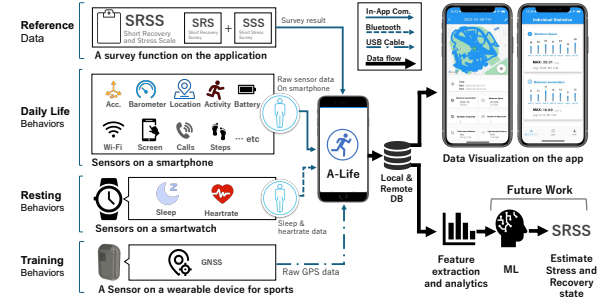


Figure 1: Data Collection and Processing Flow

four infielders, and four outfielders. We extracted daily assessed digital phenotypic variables from smartphone and wearable devices, including Apple Watch and GPS device data. Additionally, The experiment was approved by our university's Ethics Committee.

3.1 Reference Data: Mental and Physical State

As reference data for the mental and physical states of SAs, we used SRSS subscales, which measure stress and the recovery status through eight questions specifically designed for athletes. Participants completed the SRSS every morning at around 8 AM using a data collection application.

The SRSS consists of two parts: The Recovery dimension (*Physical Performance, Mental Performance, Emotional Balance, and Overall Recovery*) and the Stress dimension (*Muscular Stress, Lack of Activation, Negative Emotional State, and Overall Stress*), encompassing a total of eight scales. Respondents rate their recovery and stress on a Likert scale from 0 to 6 compared to past experiences. Higher SRS scores indicate better recovery, while higher SSS scores indicate more stress.

3.2 Daily Activity Data from Smartphone and Wearable Device

In this study, 12 types of data collected from smartphones, smartwatches, and wearable GPS devices for sports were used to gather daily activity data of SAs, which were categorized into three types: **daily life, resting, and training**.

For daily life activities, data were passively collected from smartphones using the AWARE Framework [21]. Data collected on a smartphone included acceleration, barometer, location, step count, activity recognition, battery, Wi-Fi, screen, and call. Sensors operated continuously: acceleration at 10 Hz, barometer at 1 Hz, and the others every 3 minutes. Battery, screen, and call data were event-based.

In this study, the sleep measurement function of the Apple Watch [14] was used to collect resting activity data. The measurement results included total sleep duration and the ability to record light sleep, deep sleep, core sleep, and awake states; corresponding to stages N1+N2, N3, rapid eye movement (REM), and awake, as defined by the American Academy of Sleep Medicine scoring model. Location data was used during training sessions to measure the workload of each training activity. This data was collected through a GPS device called Digitalyst Type S [18], designed for

sports training. This GPS device collected location data at 10 Hz and exported it directly to the smartphone via a cable. The location data included latitude, longitude, altitude, and acceleration. Athletes were required to wear the GPS device on their back using GPS innerwear and to turn it on during every training session whenever possible.

3.3 Feature Extractions and Data Analysis Methods

In this study, various statistical features were extracted from the data collected during daily activities, resting periods, and training sessions. These features included the mean, variance, total, maximum, minimum, and median values for each date and designated time period. The day was segmented into four distinct periods, morning (0–5 o'clock), afternoon (6–11 o'clock), evening (12–17 o'clock), and night (18–24 o'clock). Latitude and longitude information from the location sensors was used to extract the total distance, speed, and locations of stays. The density-based spatial clustering of applications and the noise [11] algorithm were employed to detect visited places from location history, calculating the time spent at each visited place, the number of visited places, and the distance from home. During training sessions, location data was used to calculate movement speed and distance. Additionally, sprinting and high-intensity sprinting events were identified when a player's speed exceeded 18 km/h and 21 km/h, respectively, for at least 0.5 seconds.

The data extracted from each sensor on the smartphone, wearable data, and scores on the SRSS subscales were synchronized before starting the analysis. Statistical analysis was then performed on the behavioral data collected from multiple sensors and various SRSS indicators.

3.4 Remuneration

Participants will receive 1000 yen per week. If data collection exceeds 70% of expectations, an additional reward will be given. A 500 yen deduction will apply for not uploading weekly sensor data or incomplete questionnaires.

4 RESULTS

4.1 Glance of Dataset

Participants provided a total of 285 SRSS responses, averaging 15.0 responses per participant (range: 1–30; SD: 8.94). Behavioral data from smartphones averaged 25.84 days per participant, with step and activity recognition data collected for 23.16 days, and other metrics (for example, smartphone usage, acceleration, barometric pressure, location, battery, screen activity, and Wi-Fi) for 27.16 days. Sleep data was recorded over 165 nights from 12 participants, and location data for training were collected 122 times from 14 participants.

4.2 Data Collection Rate and Subjective Load

Fig. 2 shows the data recovery rates of each sensor. Smartphone sensors, such as battery (92.81%), steps (90.19%), and location (89.47%) exhibit the highest and most stable recovery rates. Conversely, the data recovery rates for the Apple Watch and GPS devices (47.92%,

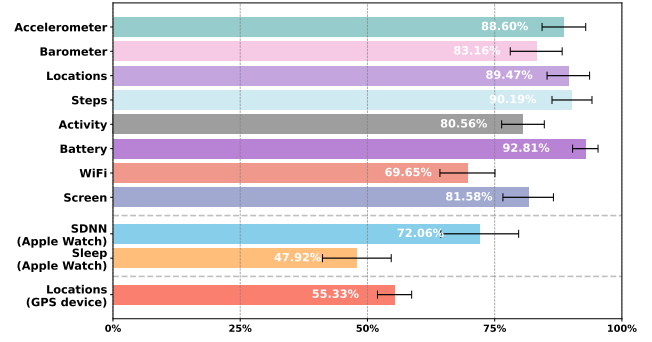


Figure 2: Data Recovery Rates of Each Sensor

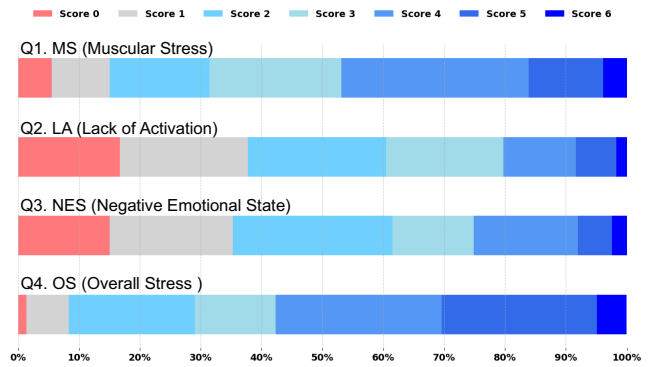


Figure 3: Survey Responses of SSS

55.33%) are lower, probably because the SAs are less accustomed to wearing these devices.

Fig. 3 shows the distribution of response rates for each score in the SSS answers of the SRSS. In the SSS questionnaire, the MS item most commonly received scores of 4 (78 occurrences) and 5 (73 occurrences). For LA and NES items, the most common responses were scores of 2, with 75 and 65 occurrences, respectively. Overall, for the OS item, the most frequent response was a score of 4, with 88 occurrences. This suggests that during practice, SAs experience high levels of physical and overall stress, while psychological stress indicators, such as LA and NES show lower stress levels.

4.3 Analysis of Behavioral Specifics

The sensor data from daily life, resting, and training, which are closely linked to the recovery of physical and mental states, were further analyzed.

Statistical Analysis: 1) *Lifestyle data*, such as step count, call duration, and distance traveled were further analyzed. The average step count was 7978.34 (SD: 4041.23), and the average number of steps peaks at around 6 AM, which is attributed to early morning practices during the measurement period, with many SAs traveling to the practice field before this time. 2) *The sleep data* included 160 samples where sleep duration was between 3 and 12 hours. The average bedtime was 24:06 (SD: 1.82) hours, and the average wake-up time was 6:00 (SD: 0.49) hours. The average duration of core

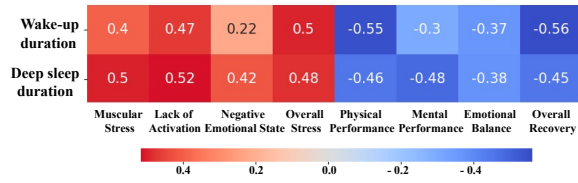


Figure 4: Pitcher's Sleep Data and SRSS Correlation Scores

sleep was 24.57 minutes (SD: 7.23), deep sleep was 16.38 minutes (SD: 20.01), REM sleep was 11.93 minutes (SD: 15.40), and wake-up time was 15.98 minutes (SD: 4.81).

Due to noise, only position data with measurement times exceeding 1 hour and travel distances over 1 km were analyzed. The final dataset included 14 players (3 infielders, 3 outfielders, 4 pitchers, and 3 catchers) over 90 sessions. The average measurement time was 3.25 hours (SD: 2.08), and the average travel distance was 7117.8 meters (SD: 3916.6), with 1284.7 meters (SD: 1730.1) in high sprint states. Errors in travel distance could occur due to the 10 Hz measurement frequency, resulting in longer than actual distances. The maximum movement speed and acceleration were 24.57 km/h (SD: 7.73) and 16.43 m/s² (SD: 4.81), respectively.

Given these findings, and recognizing the unique characteristics of baseball positions, SAs were classified by positional attributes for correlation and statistical analysis. The average number of sprints was: Pitcher 20.22 (SD: 21.86), Outfielder 20.08 (SD: 15.45), Catcher 23.62 (SD: 28.88), and Infielder 13.26 (SD: 15.69). The average distances covered were: Pitcher (7329.01m), Outfielder (8168.65m), Catcher (10034.09m), and Infielder (9301.55m). This indicates that SAs in different positions experience varying physical loads, highlighting the need for tailored physiological and psychological monitoring based on specific positions.

Correlation Analysis: We comprehensively analyze the correlation among the collected data. Due to space limitations, we merely show one case. For example, the correlation between sleep data and the physical&mental states of SAs was analyzed (i.e., Spearman correlation) (Fig. 4). The results demonstrated that wake-up duration is positively correlated with overall stress ($r = 0.50$, $p < 0.05$) and negatively correlated with overall recovery ($r = -0.56$, $p < 0.05$). Additionally, deep sleep duration is positively correlated with overall stress ($r = 0.48$, $p < 0.05$) and negatively correlated with overall recovery ($r = -0.45$, $p < 0.05$). These findings empirically confirm the strong correlation between sleep and the physical and mental states of SA. Furthermore, they demonstrate the practicality and effectiveness of the developed system.

5 DISCUSSION

This study found that passive sensing systems can better understand the psychological stress of SAs; additionally, it automatically collect data on "exercise intensity during competitions" and "lifestyle patterns." Significant differences in physical load and performance among athletes in different positions were revealed, highlighting the need for personalized monitoring mechanisms. Data for this study was collected during the university's spring break season, coinciding with the team's spring training period. Therefore, the

daily behavior patterns of SAs may differ during the academic semester. Additionally, data were collected from only one baseball team over four weeks and involved just 19 participants. To improve the validity of the research, future studies should increase the sample size of SAs and apply the proposed method to more various subjects over longer periods.

This paper examined sensor data from mobile and wearable devices, focusing on physical and mental stress in SAs. The results indicate that this data can serve as digital phenotyping to study SAs' behavioral patterns. Based on these findings, the next step is to develop a machine-learning model to detect SA conditions using these features. Future applications include advancing personalized mental health interventions and intelligent health monitoring systems for athletes.

6 CONCLUSION

This paper details the design and implementation of a foundational system for detecting the physical and mental states of SAs using passive mobile sensing. Preliminary results show that passive sensing can automatically collect data on "daily life patterns" and "exercise intensity", which are both closely related to recovery. Future research will focus on data collection during academic and competitive periods and employ machine learning for analysis and prediction. This study aims to catalyze further research into SAs' health and well-being, driving advancements in SportsHCI.

ACKNOWLEDGMENTS

This research was partially funded by JSPS KAKENHI Grant (23K17004).

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